WASP-12b

R-1400

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-12_200103` · created 2026-08-01T04:02:52+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458851.73631 +/- 0.00068 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.044
Transit depth (Rp/Rs)^2	1.8e-05 +/- 0.0037 [%]
Orbital Inclination (inc)	82.0 +/- 2.88
Airmass coefficient 1 (a1)	1.22 +/- 0.74
Airmass coefficient 2 (a2)	-0.15 +/- 0.8
Scatter in the residuals of the lightcurve fit is	61.55 %
Best Comparison Star	None
Optimal Aperture	0.75
Optimal Annulus	3.75
Transit Duration (day)	0.1088 +/- 0.009

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.0 ± 0.044 vs 0.1178 (deviation 2.68σ)
Duration fit vs published	0.1088 d vs 0.125 d (deviation 1.8σ)
Mid-time O–C	1.1 min
Depth SNR	0.0
Residual scatter	61.55 %

Light curve

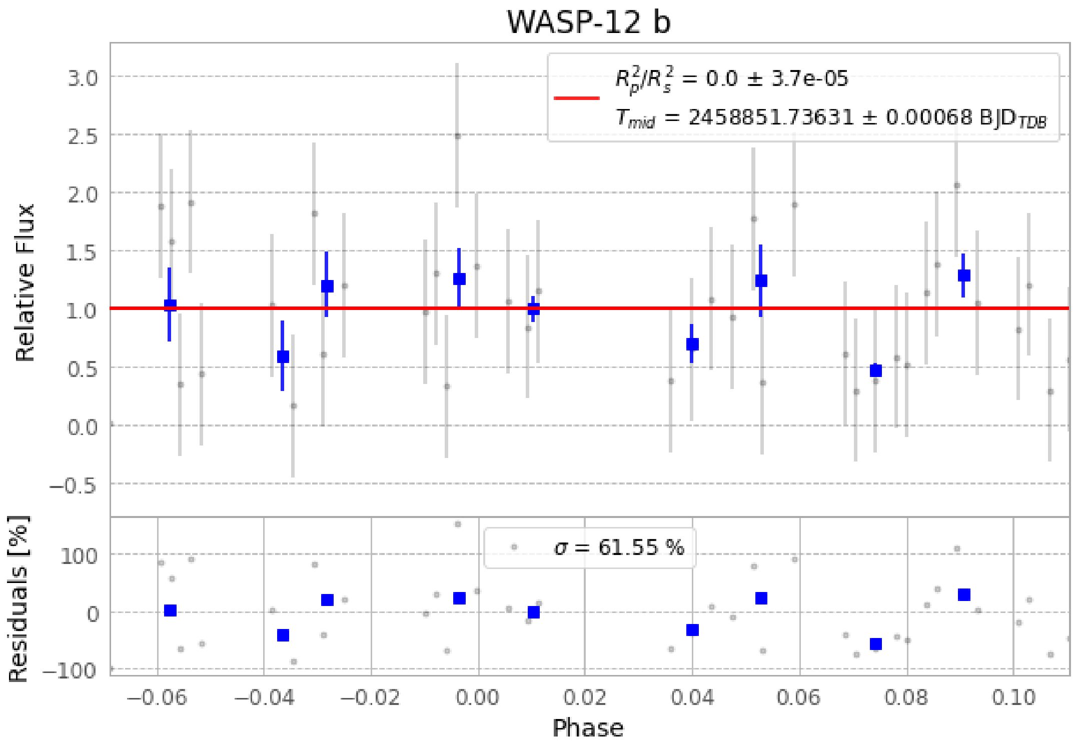

Figure 1 — FinalLightCurve_WASP-12 b_2020-01-02.png (SHA-256 bd93a0e3e0ed1f03…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [372, 301], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 2a40fc0acf241e83…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

101 FITS frames (66 MB), `WASP-12200103032412.FITS` → `WASP-12200103082412.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-200103.log (e9fd47853af5…), FinalLightCurve_WASP-12 b_2020-01-02.png (bd93a0e3e0ed…), FinalLightCurve_WASP-12 b_2020-01-02.csv (3422d1a78cae…)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-08-01 04:02Z.